SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	M Jeevitha	**Reg. No.:**	192424257
Section / Batch:	2024-C	**Date:**	25-8-2026

Code of Conduct

*I **M Jeevitha (192424257**)Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:**Jeevitha**

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression Possible Antecedents Valid/Invalid Reason

d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. **Maintain entity coherence using assignment, mistakes, and you.**
2. **Use a Cause–Effect or Elaboration discourse relation.**
3. **Include at least two keywords from:**
    - **review**
    - **improve**
    - **confident**
    - **feedback**
4. **Response length must be 2–3 sentences.**

5. **Maintain a positive and polite tone.**
6. **Do not give contradictory advice.**

Tasks:

a) Identify the dialog acts required.

b) Identify the important entities that must be maintained.

c) Generate three possible responses satisfying all constraints.

d) Compare the responses based on:

- **Coherence**
- **Constraint satisfaction**
- **Politeness**
- **Relevance**

e) Select the best response and justify your answer.

f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- **A financial institution**
- **The side of a river**

Constraints:

1. **Resolve the correct word sense using contextual information.**
2. **Identify important entities and actions.**
3. **Represent the meaning using predicate logic.**
4. **Preserve semantic relationships.**
5. **Maintain coherence in the generated paraphrase.**

Tasks:

a) Identify the possible meanings of bank.

b) Select the correct sense and justify your decision using contextual constraints.

c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	7	7	7	6	6	33	94
2	7	7	7	6	6	33	
3	6	6	6	5	5	28	

Faculty In-charge	Course Coordinator	Program Director
Dr. T. Kumaragurubaran		
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1. Reference Resolution and Discourse Coherence

Given paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

a) Referring expressions and possible antecedents

Referring Expression	Possible Antecedents
He	Ravi, Arun
him	Ravi, Arun
it	Book on Artificial Intelligence
they	Ravi and Arun
the book	Book on Artificial Intelligence

b & c) Applying constraints

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He	Ravi	**Valid**	Ravi is a singular male; grammatical agreement is satisfied.
He	Arun	**Valid**	Arun is also a singular male, so gender/number agreement is satisfied.
him	Ravi	**Valid**	Ravi is male and singular; semantically, Ravi can be helped.
him	Arun	**Invalid**	Arun is the person performing the helping action, so referring to himself as "him" is less coherent.
it	Book	**Valid**	"It" is singular and refers naturally to the book.
it	Ravi/Arun	**Invalid**	People are normally referred to using "him/he," not "it."
they	Ravi + Arun	**Valid**	"They" is plural and refers naturally to both people.
the book	AI book	**Valid**	Recency and semantic context indicate the previously mentioned book.

Important point about "He"

"He" is **ambiguous by gender and number alone**, because both Ravi and Arun are male and singular.

However, **recency** favors **Arun**, because Arun is the most recently mentioned person before "He."

So:

He → Arun is the most likely resolution.

d) Final coreference chains

The most coherent interpretation is:

- **Arun → He → him**
- **Ravi → they**
- **Arun → they**
- **Book on Artificial Intelligence → it → the book**

So the chains are:

Chain 1: Arun → He → him

Chain 2: Ravi + Arun → they

Chain 3: Book on Artificial Intelligence → it → the book

Note: “He” is technically ambiguous, but under the given constraints, **Arun** is the preferred antecedent because of recency and discourse coherence.

e) Rewrite the paragraph with references

Rewritten paragraph:

Ravi met Arun at the library. **Arun** was looking for a book on Artificial Intelligence. Arun helped **Ravi** find the book. Later, **Ravi and Arun** discussed the book before leaving.

This version removes all pronoun ambiguity.

f) Ambiguity if “He was looking for a book” appears before Arun is introduced

If the sentence were:

“Ravi met at the library. He was looking for a book. Arun arrived later.”

or if **“He was looking for a book”** appeared before Arun was introduced, then **He** could only refer to Ravi if Ravi had already been introduced.

If neither Ravi nor Arun had been introduced, **“He” would have no clear antecedent**.

Therefore, reference resolution requires a previously introduced entity and sufficient contextual information.

Short exam answer:

If “He was looking for a book” occurs before Arun is introduced, “He” cannot refer to Arun because Arun is not yet available as an antecedent. This may cause ambiguity or an unresolved reference. The system would rely on discourse context to determine the intended person.

2. Dialog Act Recognition and Response Generation

Given conversation:

Student: “I submitted my assignment, but I am worried that I may have made many mistakes.”

a) Required dialog acts

The required dialog acts are:

1. **Reassure** – reduce the student's worry.
2. **Advise** – provide useful guidance about the assignment.

So:

Dialog Act = Reassure + Advise

b) Important entities

The important entities that should remain coherent are:

- **assignment**
- **mistakes**
- **you**

The response should continue talking about the student's **assignment** and possible **mistakes**, rather than suddenly changing the topic.

c) Three possible responses

Response 1

Don't worry, you have already completed the assignment, and mistakes are a normal part of learning. You can **review** the assignment and use **feedback** to **improve** your work and feel more **confident** next time.

Response 2

It is completely normal to worry about mistakes after submitting an assignment, so you should not be too hard on yourself. When you receive **feedback**, **review** your mistakes carefully so you can **improve** and become more **confident**.

Response 3

You have done your part by submitting the assignment, so try to stay **confident**. If there are mistakes, use the **feedback** to **review** them and **improve** your future assignments.

d) Comparison of responses

Criteria	Response 1	Response 2	Response 3
Coherence	Excellent	Excellent	Excellent
Constraint satisfaction	Excellent	Excellent	Excellent
Politeness	Positive and polite	Positive and supportive	Positive and supportive
Relevance	Highly relevant	Highly relevant	Highly relevant
Advice	Review + feedback	Review + feedback	Review + feedback

e) Best response

Best choice: Response 2

It is completely normal to worry about mistakes after submitting an assignment, so you should not be too hard on yourself. When you receive **feedback**, **review** your mistakes carefully so you can **improve** and become more **confident**.

Justification:

Response 2 is the best because:

- It **reassures** the student.
- It provides clear **advice**.
- It maintains entity coherence using **assignment/mistakes/you**.

- It contains the required keywords **feedback, review, improve, confident**.
- It has a clear **cause–effect relation**: feedback → review mistakes → improve → become confident.
- It is positive, polite, and relevant.
- It does not give contradictory advice.

f) What happens if entity coherence is not maintained?

If entity coherence is not maintained, the response may become **confusing or irrelevant**.

For example:

“Don't worry about your assignment. You should review your **car** and use feedback to improve your **project presentation**.”

Here, the topic suddenly changes from the **assignment and mistakes** to a car and presentation. This breaks discourse coherence and makes the response difficult to understand.

Exam answer:

When entity coherence is not maintained, the system may lose track of the main topic or entities. This can produce irrelevant, confusing, or logically inconsistent responses.

3. Word Sense Disambiguation and Meaning Representation

Source sentence:

“Priya sat on the bank and watched the boats moving across the water.”

a) Possible meanings of “bank”

The word **bank** has two possible meanings:

1. **Financial institution** – a place where people deposit or withdraw money.
2. **Side of a river –** the land beside a river or body of water.

b) Correct sense and justification

Correct sense:

Bank = side of a river

Why?

The surrounding words provide strong contextual clues:

- **sat on the bank**
- **boats**
- **water**
- **moving across the water**

These words are strongly associated with a **riverbank**.

A financial bank does not normally have boats moving across water beside it.

Therefore:

bank = riverbank / side of a river

This is an example of **Word Sense Disambiguation (WSD)**, where the correct meaning of a word is selected using its surrounding context.

c) Predicate logic representation

We can represent the sentence as:

sit(Priya, bank)

bank(bank)

beside(Priya, bank)

watch(Priya, boats)

boat(boats)

move(boats)

water(water)

A more complete representation is:

$$\exists x \exists y \exists z \exists w$$
$$Priya(x) \land bank(y) \land sit(x,y) \land beside(x,y)$$
$$\land\, boat(z) \land water(w) \land watch(x,z)$$
$$\land\, move(z) \land across(z,w)$$

Meaning:

- **Priya(x)** → x is Priya
- **bank(y)** → y is a riverbank
- **sit(x,y)** → Priya sits on the bank
- **boat(z)** → z is a boat
- **watch(x,z)** → Priya watches the boats
- **move(z)** → boats are moving
- **water(w)** → w is water
- **across(z,w)** → boats move across the water

d) Simple English paraphrase

Priya sat beside the river and watched the boats moving across the water.

This removes the ambiguity of the word **bank**.

e) How WSD improves Machine Translation

Word Sense Disambiguation helps Machine Translation by selecting the correct meaning of an ambiguous word before translating it.

For example:

"Priya sat on the bank."

If **bank** is incorrectly interpreted as a financial institution, the translation may produce a wrong meaning.

After WSD:

bank → riverbank

the translation system can produce a sentence meaning:

“Priya sat beside the river.”